SUM INSTEAD OF AND: WHY DETECTORS TRAINED ON NORMAL-VS-FAULT DATA FLAG LOOK-ALIKES, AND THE MINIMAL SUPERVISION THAT FIXES IT

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ABSTRACT

A detector trained to separate normal operation from one target condition in multivariate time series can reach near-perfect accuracy and still raise many false alarms on other conditions that reproduce only part of the target's signature. We show that when the target is a *conjunction*, k sensors shifting together, the cause is not a spurious feature but the logical form of the learned rule. For the normal-vs-target training task the Neyman–Pearson-optimal detector is additive across sensors, so more data and class rebalancing converge to it. In the well-separated regime, any norm-limited network places near-miss patterns close to the target, and pushing all k of them back to the normal score requires at least k times the minimum path norm. The additive rule obeys a closed-form false-alarm law that predicts how often each partial look-alike is flagged from the detector's own detection rate, and implies a *detectability paradox*: the better the target is detected, the more look-alikes are flagged. Deployment is a test against a composite null of all partial look-alikes. Against look-alikes of unbounded strength, additive and uncapped channel-independent detectors have no valid threshold with nonzero power, while the intersection-union test, which capped-evidence detectors can represent, is valid and misses at most k times as often as the best valid detector. We characterize the supervision that identifies the conjunction: k channel near-misses and, by Helly's theorem on the line, $\binom{k}{2}$ pair-staggered temporal near-misses, at a per-type sample cost governed by the weakest sensor. This predicts that "one after another" negatives, a natural time augmentation, do not teach simultaneity. Multi-seed synthetic experiments confirm the law, the separated-regime bound and the timing prediction. On counterfactually spliced fault windows from public building-system simulations, pair-staggered negatives teach simultaneity where fully staggered ones do not, and channel near-misses cut fan-coil look-alike alarms, including on never-seen faults.

1 INTRODUCTION

Many monitoring problems ask whether a specific condition is present in a multivariate time series: a failing capacitor in a heat pump, a stuck valve in an air-handling unit, a fault in a chemical process. The standard recipe trains a classifier on windows labelled *normal* or *target condition*, sets an alarm threshold for a small false-alarm rate on normal data, and deploys (Katipamula & Brambley, 2005; Zhao et al., 2019; Venkatasubramanian et al., 2003). In deployment the detector also sees many conditions it was never trained on. Some of them move a *subset* of the sensors that define the target. A detector that is excellent on the training task can then become a source of false alarms. On a public building benchmark (section 6), an LSTM trained on one fan-coil fault detects it in 100% of windows at a 1% false-alarm budget. Yet it flags 63–79% of windows from other, never-seen faults that move one or two of the fault's three signature sensors, against 13–36% for a hand-written rule requiring all three.

The question. Why does training produce a rule that is broader than the condition, and what is the least supervision that fixes it? Prior work on shortcut learning explains failures in which a model relies on a *spurious* feature correlated with the label (Geirhos et al., 2020; Shah et al., 2020;

Hermann et al., 2024). Here no spurious feature is needed. The model uses the right sensors and combines them with the wrong logic: a *sum* of per-sensor evidence instead of an *and*. Throughout, picture three warning lights that always switch on together for the target. Training shows only "all off" (normal) and "all on" (target). For that data, counting lit lights is as good as requiring all three. A condition that lights two of them is never shown, so nothing in training penalizes calling it the target.

Contributions.

1. **The broad rule is the training optimum (theorem 1).** Under a Gaussian conjunctive-signature model, the Bayes and Neyman–Pearson (NP) detector for normal vs. target is linear, hence additive across independent sensors. It does not depend on class prevalence, and every cross-entropy-consistent learner converges to it. More data and rebalancing therefore cannot remove the failure.
2. **A closed-form false-alarm law and a detectability paradox (theorem 2 and corollary 3).** Any other condition is flagged with probability $\Phi(D\, r - z_\alpha)$, where r is the share of the target's signal it reproduces. Written in terms of the detection rate β, look-alike false alarms increase with β and tend to one as $\beta \to 1$.
3. **Deployment as a composite null (theorem 4).** Against all partial look-alikes, additive detectors lose a $\sqrt{k}$ factor in effective signal, or all power when look-alikes can be arbitrarily strong. This includes channel-independent models with unbounded per-channel terms, a design common in time-series foundation models (Nie et al., 2023; Goswami et al., 2024). The intersection-union test, which capped-evidence detectors can represent, is valid, and its miss probability is within a factor k of the smallest possible.
4. **Why norm-seeking training prefers the sum (theorem 5).** When classes are separated, any network with Lipschitz constant κ times the minimum places each near-miss at least $1 - \kappa/\sqrt{k}$ of the way to the target. Rejecting all k near-misses costs at least k times the minimum path norm.
5. **Minimal supervision (theorem 6 and proposition 7).** k channel near-misses and $\binom{k}{2}$ pair-staggered temporal near-misses are necessary and sufficient to identify the rule within per-axis monotone classes. The temporal count follows from Helly's theorem on the line, which also implies that fully staggered negatives do not identify simultaneity. A per-type sample bound $32(\sigma/\Delta)^2 \log(4k/\eta)$ makes the weakest sensor the bottleneck.
6. **Evidence.** Multi-seed synthetic experiments match the law to within 0.01 for logistic regression and 0.03 for a two-layer net, and confirm the separated-regime bound and the Helly prediction. In a toy replica, generic outlier negatives leave look-alike alarms unchanged while near-misses remove them. On counterfactually spliced faults from public building-system simulations (Granderson et al., 2023), the timing prediction holds on all three fan-coil runs, and look-alike alarms rise with detection where detection is not saturated.

Proofs are in appendix A. Experimental details and further results are in appendices B and C.

2 Related work

Shortcut learning and simplicity bias. Networks often rely on easy, label-correlated features that fail out of distribution (Geirhos et al., 2020; Shah et al., 2020). Mechanisms include gradient starvation (Pezeshki et al., 2021), feature availability and predictivity (Hermann & Lampinen, 2020; Hermann et al., 2024), parameter-space volume (Scimeca et al., 2022), representation biases (Lampinen et al., 2024), low-dimensional projections in one-hidden-layer nets (Morwani et al., 2023; Tsoy & Konstantinov, 2024) and NTK eigenstructure (Lim et al., 2025). Underspecification shows that many predictors fit the training data and differ out of distribution (D'Amour et al., 2022). These analyses contrast one *core* with one *spurious* feature. In our setting every feature is causal. What is underdetermined is how the features combine, and the out-of-distribution inputs are partial look-alikes of the target rather than background shifts.

Implicit bias and norms. Gradient descent on separable data converges in direction to max-margin or KKT solutions for linear and homogeneous models (Soudry et al., 2018; Lyu & Li, 2020; Ji & Telgarsky, 2020; Chizat & Bach, 2020). Small-initialization two-layer leaky-ReLU nets on

linearly separable symmetric data converge to the *linear* max-margin classifier (Lyu et al., 2021). Weight decay acts as a path- or representation-cost penalty (Neyshabur et al., 2015; Savarese et al., 2019; Ongie et al., 2020). We use these facts only through a norm-to-Lipschitz bound. This yields statements that hold for any trained network in terms of a measurable constant (theorem 5), rather than a convergence theorem.

Learning interactions. Parities and high-order sparse functions are hard for gradient methods (Barak et al., 2022; Abbe et al., 2022), and interaction detection from trained weights is studied in Tsang et al. (2018). Additive models are a classical interpretable class (Hastie & Tibshirani, 1990; Agarwal et al., 2021). Our conjunction is low-order and easy to represent. The obstacle is identifiability from normal-vs-target labels, not computation.

Robust training and counterfactual supervision. Group-robust methods need group labels or error-based proxies (Sagawa et al., 2020; Liu et al., 2021; Kirichenko et al., 2023), and invariance methods need environments (Arjovsky et al., 2019). When the training error is zero, as in our separated regime, error-upweighting has nothing to upweight. Counterfactual or contrast examples and hard negatives improve local decision boundaries (Kaushik et al., 2020; Gardner et al., 2020; Teney et al., 2020; Robinson et al., 2021). We give a theory of *which* contrasts identify a conjunction and how many are needed. Teaching sets (Goldman & Kearns, 1995; Zilles et al., 2011) and membership queries for conjunctions (Angluin, 1988) are the classical analogues for the channel part. To our knowledge the temporal part (theorem 6(b)) is new.

Open-set and out-of-distribution detection. Open-set recognition and OOD detection flag inputs unlike the training classes (Scheirer et al., 2013; Hendrycks & Gimpel, 2017; Yang et al., 2024). Outlier exposure trains on generic auxiliary outliers (Hendrycks et al., 2019), and good closed-set classifiers tend to be good open-set detectors (Vaze et al., 2022). Our paradox concerns a different population: look-alikes that share the target's features. For those, better closed-set detection provably *increases* false alarms, and generic outliers carry no identifying information (section 6).

Multivariate time series and fault detection. Deep detectors for multivariate time series include recurrent, graph and transformer models (Hundman et al., 2018; Su et al., 2019; Zhao et al., 2020; Deng & Hooi, 2021; Xu et al., 2022; Tuli et al., 2022), and benchmark practice has been criticized (Wu & Keogh, 2023; Kim et al., 2022; Schmidl et al., 2022). Channel independence trades capacity for robustness in forecasting (Nie et al., 2023; Han et al., 2024) and underlies several foundation models (Goswami et al., 2024; Das et al., 2024; Ansari et al., 2024). Multi-sensor change detection targets changes in an unknown *subset* of sensors (Xie & Siegmund, 2013; Page, 1954), which is the opposite of requiring all k. Building and process fault detection and diagnosis (Katipamula & Brambley, 2005; Kim & Katipamula, 2018; Zhao et al., 2019; Downs & Vogel, 1993) motivate our setting. The statistical tools we use are NP testing (Neyman & Pearson, 1933; Lehmann & Romano, 2005), intersection-union tests (Berger, 1982; Berger & Hsu, 1996) and Helly's theorem (Helly, 1923).

3 Problem formulation

A window is $X \in \mathbb{R}^{T\times C}$ (T steps, C sensors). We write $x = \mathrm{vec}(X) \in \mathbb{R}^d$ with $d = TC$. The target condition has a *signature* $S \subseteq [C]$, $|S| = k \geq 2$, with directions $s_j \in \{\pm 1\}$, shifts $\delta_j > 0$ and active intervals $I_j \subseteq [T]$:

$$X = W + G, \qquad G_{tj} = s_j \delta_j \mathbf{1}[t \in I_j]\, \mathbf{1}[j \in S], \qquad g = \mathrm{vec}(G), \tag{1}$$

where W is the normal process. The target is *simultaneous* if all I_j equal one interval I. Other conditions are modelled by other patterns $h \in \mathbb{R}^d$. The ones that matter are *partial* patterns that keep the target's shift on a strict subset $A \subsetneq S$ (*channel near-misses* when $|A| = k - 1$), and patterns that shift all of S with non-simultaneous intervals (*temporal near-misses*).

Assumption 1 (Gaussian location model)**.** $W \sim \mathcal{N}(0, \Sigma)$ with $\Sigma \succ 0$. Condition h produces $x \sim \mathcal{N}(h, \Sigma)$; normal is $h = 0$ and the target is $h = g$.

We write Φ for the standard normal CDF and $z_\alpha = \Phi^{-1}(1-\alpha)$, and define $D^2 = g^\top \Sigma^{-1} g$ (target detectability) and $r(h) = g^\top \Sigma^{-1} h / D^2$ (the share of the target's signal that h reproduces). If sensors

are independent (Σ block-diagonal across channels) and $h = g \odot \mathbf{1}_A$, then $r = \sum_{j \in A} D_j^2 / D^2$ with $D_j^2 = g_j^\top \Sigma_{jj}^{-1} g_j$. A detector ϕ is α-*calibrated* if it flags a fraction α of normal windows. This matches standard practice and all our experiments.

Two questions, two nulls. Training data pose the *simple* test $H_0 : h = 0$ vs. $H_1 : h = g$. Deployment poses a *composite* test: the target vs. normal *or* any look-alike. For a shift statistic $a \in \mathbb{R}^k$ on the signature sensors (section 4.2) we write the overlap null as $\mathcal{N}(M) = \{\mathcal{N}(\mu, \Sigma_S) : \mathrm{supp}(\mu) \subsetneq S,\ 0 \le \mu_j \le M_j\}$. A detector is *deployment-valid* at level α if its false-alarm rate is at most α under normal and under every member of $\mathcal{N}(M)$.

4 Theory

4.1 The training optimum is additive, and what it does to look-alikes

Theorem 1 (Training-optimal detector)**.** *Under Assumption 1 with training prevalence $\pi \in (0,1)$: (i) the Bayes log-odds is $\ell_\pi(x) = g^\top \Sigma^{-1} x - \frac{1}{2} D^2 + \log \frac{\pi}{1-\pi}$, linear in x and a sum of per-sensor terms when sensors are independent; (ii) for every α, the most powerful size-α test of normal vs. target is $\phi^*(x) = \mathbf{1}\{g^\top \Sigma^{-1} x > z_\alpha D\}$, which does not depend on π; (iii) if a learner's score $\hat{f}_n \to \ell_\pi$ in probability under the training mixture and $\hat{\phi}_n$ thresholds it at the $(1-\alpha)$-quantile of its scores on normal data (the population quantile, or the empirical one from $m_n \to \infty$ normal windows independent of training), then $Q(\hat{\phi}_n \neq \phi^*) \to 0$ for every distribution Q absolutely continuous w.r.t. Lebesgue measure.*

Part (iii) applies to well-specified logistic regression and to any cross-entropy-consistent flexible learner. In this regime the broad rule is not a failure of optimization. It is the unique optimum of the training problem, so additional normal-vs-target data or rebalancing moves a learner *toward* it.

Theorem 2 (Partial-overlap false-alarm law)**.** *Under Assumption 1, $\Pr_h(\phi^* = 1) = \Phi\big(D\, r(h) - z_\alpha\big)$ for every pattern h. With independent, equal-SNR sensors, an m-of-k partial pattern is flagged with probability $\Phi(\frac{m}{k} D - z_\alpha)$. A pattern that shifts a single signature sensor by c times its target shift is flagged with $\Phi(\frac{c}{k} D - z_\alpha)$, a sign-flipped target with $\Phi(-D - z_\alpha)$, and a shift on non-signature sensors at rate exactly α.*

Corollary 3 (Power–specificity curve; detectability paradox)**.** *Let $\beta = \Phi(D - z_\alpha)$ be the detection rate. Then $\Pr_h(\phi^* = 1) = \Phi\big(r(h)\,(\Phi^{-1}(\beta) + z_\alpha) - z_\alpha\big)$. For any look-alike with $r(h) \in (0,1)$ this is strictly increasing in β and tends to 1 as $\beta \to 1$. Anything that makes the target easier to detect at a fixed false-alarm budget while leaving the look-alike's share $r(h)$ unchanged raises its false-alarm rate. Scaling all shifts (of target and look-alike) up, or the noise down, does this for* every *partial look-alike.*

The curve has no free parameter. Given a model's measured detection rate it predicts its look-alike rates, and agreement is evidence that the model behaves like a sum of evidence. The law also shows which look-alikes are worst. A condition that strongly moves the *loudest* signature sensor (largest D_j) is flagged most, and a strongly amplified single sensor can exceed the target itself ($r > 1$).

4.2 What deployment needs: a test against all look-alikes

Let a_j be the post-onset minus baseline mean of sensor $j \in S$, oriented by s_j. Under Assumption 1 with independent sensors and a known onset, $a \sim \mathcal{N}(\mu, \mathrm{diag}(\sigma_j^2))$ with $\mu = (\delta_j)_{j \in S}$ under the target. A *generalized additive* detector has the form $\mathbf{1}\{\sum_j \psi_j(a_j) > t\}$ with each ψ_j nondecreasing. The intersection-union (IU) test is $\phi_{IU}(a) = \mathbf{1}\{\min_j a_j/\sigma_j > z_\alpha\}$ (Berger, 1982).

Theorem 4 (Composite null)**.** *Let $M_j \ge \delta_j$. (a) Every deployment-valid level-α detector, including any function of the full window, has power at most $\min_{j \in S} \Phi(D_j - z_\alpha)$, with $D_j = \delta_j/\sigma_j$ for the shift statistic and $D_j^2 = g_j^\top \Sigma_{jj}^{-1} g_j$ for full windows when the null contains the near-miss patterns $g - g_j$. (b) ϕ_{IU} is deployment-valid for $M = \infty$. Its miss probability is at most k times the smallest miss probability allowed by (a). (c) With equal shifts, $M_j = \delta$, a deployment-valid linear detector $\mathbf{1}\{w^\top a > t\}$ with $w \ge 0$ has power at most $\Phi(\delta/(\sigma\sqrt{k}) - z_\alpha)$, attained at $w = \mathbf{1}$. (d) If $M = \infty$ and some ψ_j is unbounded above, every deployment-valid generalized additive detector has power*

0. *This covers every linear detector with $w \geq 0$, $w \neq 0$, and every channel-independent sequence model $f(X) = \sum_j f_j(X_{\cdot j})$ whose per-channel terms are unbounded along fault-direction shifts. (e) If every ψ_j is bounded, the class contains ϕ_{IU} (take $\psi_j = \mathbf{1}\{a_j > z_\alpha \sigma_j\}$ and $t = k - \frac{1}{2}$). (f) The training optimum ϕ^* is not deployment-valid: with equal SNR, its false-alarm rate on a channel near-miss is $\Phi\big(\frac{k-1}{\sqrt{k}} \frac{\delta}{\sigma} - z_\alpha\big) > \alpha$.*

Part (d) says that capping each sensor's vote is not a cosmetic choice. Without it, no threshold makes a sum-type detector safe against look-alikes that are allowed to be loud.

4.3 Why norm-seeking training prefers the sum when classes are separated

When normal and target are far apart, as in our real data where one sensor alone separates the training set, theorem 1(iii) no longer pins down the detector away from the training support. The choice is then made by what training prefers. Gradient descent on homogeneous nets favours small-norm solutions (Lyu & Li, 2020; Ji & Telgarsky, 2020), weight decay penalizes the norm explicitly, and small norm bounds the Lipschitz constant (lemma 10).

Theorem 5 (Near-misses sit next to the target)**.** *Let x_0, x_+ be normal and target prototypes, $\Delta = \|x_+ - x_0\|$, and f a score with $G = f(x_+) - f(x_0) > 0$. Let $\kappa = \mathrm{Lip}(f)\,\Delta/G \geq 1$ be its Lipschitz constant relative to the smallest one compatible with this separation. (a) Any x with $\|x - x_+\| = \rho\Delta$ satisfies $p(x) := \frac{f(x)-f(x_0)}{G} \geq 1 - \kappa\rho$. A channel near-miss has $\rho = 1/\sqrt{k}$ (equal shifts and durations). A temporal near-miss displacing one sensor's shift block by $\Delta t \leq L$ steps has $\rho = \sqrt{2\Delta t/(kL)}$. (b) Let $f(x) = \sum_i v_i \mathrm{relu}(w_i^\top x + b_i) + c$ have path norm $R = \sum_i |v_i| \|w_i\|$. If $f(x_+) - f(x^{(j)}) \geq \gamma$ for every channel near-miss $x^{(j)}$, then $R \geq \frac{\sqrt{k}\gamma}{\delta\sqrt{L}} = k\,\frac{\gamma}{G}\,R_{\min}$, where $R_{\min} = G/\Delta$.*

A norm-limited model therefore scores near-misses most of the way to the target. Pushing them back to the normal level ($\gamma = G$) needs k times the minimum path norm. For two symmetric prototypes, the small-initialization limit of two-layer leaky-ReLU training (Lyu et al., 2021) is exactly the linear classifier $w \propto \mathbf{1}_S$ (remark 11), which places channel near-misses at $p = (k-1)/k$. Proposition 12 (appendix A) converts the norm cost into a necessary fraction of near-miss examples that grows with the weight-decay strength.

4.4 The least supervision that identifies the conjunction

A labelled set *identifies* the target rule within a hypothesis class if the target rule is the only member consistent with it. We consider the two axes along which a pattern can fall short of the target. On the *channel* axis, configurations are subsets $A \subseteq S$ of shifted sensors, all simultaneous. On the *time* axis, all of S is shifted on intervals I_j, summarized by the overlap graph $\mathcal{G}$ on S, with an edge $\{i, j\}$ iff $I_i \cap I_j \neq \emptyset$. $\mathcal{H}_{\mathrm{ch}}$ ($\mathcal{H}_{\mathrm{t}}$) is the set of rules monotone in A (in the edge set of $\mathcal{G}$) that equal 1 at the target. Normal training data label the channel-axis bottom $A = \emptyset$.

Theorem 6 (Minimal supervision)**.** *(a) A labelled set identifies $\mathbf{1}\{A = S\}$ in $\mathcal{H}_{\mathrm{ch}}$ iff it contains all k channel near-misses $S \setminus \{j\}$. (b) Intervals on a line share a common point iff every pair overlaps (Helly, 1923). Hence "simultaneous" is $\mathbf{1}\{\mathcal{G} = K_k\}$, and a labelled set identifies it in $\mathcal{H}_{\mathrm{t}}$ iff it contains, for every pair $\{a, b\}$, a* pair-staggered *pattern with $\mathcal{G} = K_k - \{a, b\}$. Each such pattern is realizable with all shifts of equal length. (c) A fully staggered pattern ($\mathcal{G} = \emptyset$, "one after another") is the bottom of the time axis. Labelling it excludes only the constant rule, so for $k \geq 3$ it neither identifies simultaneity nor replaces any pair-staggered pattern. (d) Hence identification on both axes needs and is achieved by $k + \binom{k}{2}$ near-miss types, 6 for $k = 3$.*

If a learner were monotone in the joint order of both axes, channel near-misses would lie below pair-staggered ones and $\binom{k}{2}$ types would suffice (remark 13). Learned networks need not respect that order, which is why we supply both. How many examples per type?

Proposition 7 (Examples per near-miss type)**.** *For a learner that thresholds each signature sensor and requires all of them, let sensor j have level ν_j in its near-miss and $\nu_j + \Delta_j$ in the target, with noise σ_j. If each threshold is the midpoint of the sample means of n near-misses of type j and n targets, and $n \geq 32\,\frac{\max_j \sigma_j^2}{\min_j \Delta_j^2} \log \frac{4k}{\eta}$, then with probability $\geq 1 - \eta$ every near-miss type is flagged*

Algorithm 1: Conjunction-identifying supervision (CIS)

Input: fault-free windows W, paired target effects D, onset o, signature size k, examples per type n

Estimate S: the k sensors with the largest standardized mean shift of D after onset (skip near-duplicates);

Positives: $W + \mathbf{1}_{t \geq o}\mathbf{1}_S \odot D$; negatives: normal windows (half time-matched to positives);

for $j \in S$ **do**

  add n channel near-misses $W + \mathbf{1}_{t \geq o}\mathbf{1}_{S \setminus \{j\}} \odot D$ with label 0

for *each pair* $\{a, b\} \subset S$ **do**

  add n pair-staggered near-misses with label 0: a shifted on $[o, o + \ell)$, b on $[o + \ell, o + 2\ell)$, every other $j \in S$ on $[o + \ell/2, o + 3\ell/2)$, $\ell = \lfloor (T - o)/2 \rfloor$

Train any detector; choose n from proposition 7, which depends on the weakest sensor;

with probability $\leq \Phi(-\Delta_j/(4\sigma_j))$ *and the target is missed with probability* $\leq \sum_j \Phi(-\Delta_j/(4\sigma_j))$. *If the* n *examples of a type are consecutive draws of an AR(1) sequence with coefficient* $\phi \in [0, 1)$, n *is replaced by* $n(1 - \phi)/(1 + \phi)$ *(lemma 15).*

The weakest sensor is the bottleneck: halving Δ_j quadruples the examples needed.

5 Method: conjunction-identifying supervision

The theory leaves two remedies. One is to change the inductive bias: an IU or capped-evidence detector (theorem 4(e)). The other is to keep a flexible model and supply the near-miss types of theorem 6. Both need the signature and examples of the target's effect. In many settings these come from paired runs, a faulty run and a fault-free run under identical conditions, or from simulators, so that the target's effect $D = X^{\text{fault}} - X^{\text{normal}}$ is known per window. We then build training windows by *counterfactual splicing*, $X = W + M \odot D$, with a mask $M \in \{0, 1\}^{T \times C}$ (algorithm 1). Calibration follows standard practice (1% of held-out normal windows). Theorem 4 implies that guaranteed validity against look-alikes would additionally require calibrating on near-misses, which we leave optional.

6 Experiments

Each experiment tests a prediction of section 4. We separate guarantees (theorems) from evidence (measurements) and report means $\pm$ s.d. over seeds where available. Real-data runs use one seed, and we mark them as such.

Data. *Synthetic*: Gaussian windows following eq. (1), with signatures and noise set per experiment (appendix B). *Toy replica*: 20 AR(1) channels ($\phi = 0.7$), 48 steps, a shared slow load factor on six channels, and a target that lowers three channels by 2.5σ from a random onset. *Real*: two public building-system fault benchmarks (Granderson et al., 2023): a fan-coil unit (FCU; 1 fault-free and 48 faulty one-year runs, 24 sensors after filtering) and a single-duct air-handling unit (AHU; 1 fault-free and 20 faulty runs, 26 sensors). Every faulty run is paired with the fault-free run under the same weather and schedule, so the fault effect D is known exactly. Windows are 48 steps of 15 minutes. Train, validation and test windows come from different weeks, and half of the normal training windows are time-matched to fault windows. Test constructs (Pm: m-of-k; BIG1: one sensor $\times 3$; ASYNC: fully staggered; PAIRSTAG: pair-staggered; UP: sign-flipped; OFF: moved to unrelated sensors) are counterfactual splices. In addition, every other fault file, never used in training, is a test case, bucketed by how many signature sensors it actually moves. All thresholds give 1% false alarms on held-out normal windows.

Detectors and baselines. We use logistic regression, random forests (Breiman, 2001) and gradient boosting (Friedman, 2001) on six window statistics per sensor, an MLP on the raw series, an LSTM (Hochreiter & Schmidhuber, 1997), a channel-independent (CI) model (a shared per-sensor

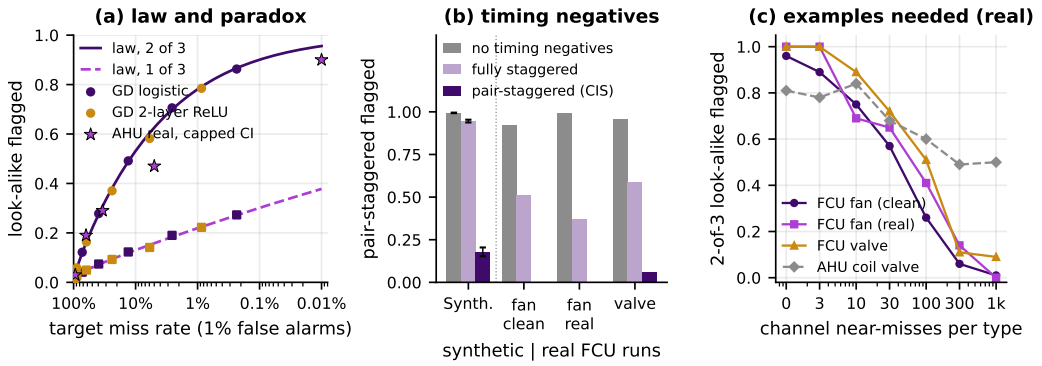


Figure 1: **(a)** The power–specificity curve of corollary 3 ($k = 3$, no fitted parameter) with measured 2-of-3 (circles) and 1-of-3 (squares) rates for gradient-descent logistic regression and a two-layer ReLU net over a signal sweep (5 seeds; s.d. smaller than markers). Stars: AHU coil-valve data, capped CI model, with the fault effect scaled by 0.05–1 (one seed); miss rates below 10^{-4} are drawn at 10^{-4}. **(b)** Temporal identification (theorem 6(b,c)): flag rate on pair-staggered look-alikes after training with no timing negatives, fully staggered negatives, or pair-staggered negatives. Synthetic: raw-series two-layer net, 5 seeds (error bars: s.d.); real: LSTM on three FCU runs (one seed each). **(c)** 2-of-3 look-alike flag rate vs. channel near-miss examples per type (LSTM, weight decay 10^{-3}, real data; detection stays ≥ 0.99).

encoder with a linear head, additive by construction, as in Nie et al., 2023) and a capped CI model ($\beta \sum_c \tanh(\cdot)$). As reference we use the rule "all signature sensors shifted in the fault direction within the same 2-hour stretch", applied to residuals after regressing out non-signature sensors. As training-time remedies we compare: more data and rebalancing; *fully staggered* negatives (a time augmentation); *generic negatives* (the fault's effect moved to unrelated sensors, an outlier-exposure-style control, Hendrycks et al., 2019); *channel near-misses*; and *CIS* (channel near-misses plus pair-staggered near-misses). Remedies add the same number of label-0 windows ($n_{\text{train}}/2$).

The law and the paradox (theorem 2 and corollary 3). For synthetic signals, the curve evaluated at each model's *own* detection rate predicts its look-alike rates. For logistic regression the largest deviation over $\delta/\sigma \in [0.5, 3]$ is 0.007 for 2-of-3 and 0.006 for 1-of-3. The two-layer net follows the same curve with largest deviation 0.023 (fig. 1a, table 3), so its learned score behaves like a sum. Detection and look-alike alarms rise together, which is the paradox. In the toy replica (table 7), the raw-series MLP, LSTM and channel-independent models lie within 0.05 of the curve at their own detection rates (0.94–0.97). Logistic regression on window statistics lies 0.06–0.07 above it and flags half of the fully staggered windows, consistent with the timing-blindness of window statistics (lemma 14). The capped CI model and the reference rule lie below the curve on 2-of-3 (0.42 vs. 0.54 and 0.21 vs. 0.52), as detectors that do not sum should. On real AHU data, scaling the fault effect raises the capped CI model's detection from 0.06 to 1.00 and its 2-of-3 alarms from 0.03 to 0.90 (stars). The stars follow the curve up to detection 0.66 and fall below it at higher detection (0.47 vs. 0.63 at detection 0.95), consistent with capping. On the FCU, detection is already 1.00 at a fault effect of $\times 0.25$, so the paradox cannot be resolved there.

Composite null and capped evidence (theorem 4). In simulation ($k = 3$), additive detectors calibrated on normal data have false-alarm rates of 0.49–1.00 on near-misses. The IU detector is valid (size 0.010), and its power $0.05/0.42/0.87/0.99$ at $\delta/\sigma = 2/3/4/5$ matches $\Phi(\delta/\sigma - z_\alpha)^3$ (table 4). Empirically, the capped CI model is far less sensitive to one loud sensor than the uncapped CI model (BIG1, no remedy: 0.09 vs. 0.99 in the toy). On real data, however, it did not learn the conjunction even with 1000 near-misses per type (FCU 2-of-3 rate 0.99). Representability (theorem 4(e)) is not learnability.

Separated regime (theorem 5). Two-layer ReLU nets trained on well-separated normal and target clusters ($k = 2, \ldots, 8$; table 5) place channel near-misses at $p = 0.57$–0.91 of the way to the target. This is slightly above the linear limit $(k-1)/k$ and above the bound of theorem 5(a) evaluated with

Table 1: Toy replica: flag rates at 1% false alarms (mean ± s.d., 3 seeds) for each training remedy, LSTM. All remedies add the same number of label-0 windows. The fault effect moved to unrelated sensors (OFF) is flagged at ≤ 0.02 under every remedy.

Training	Detection	1-of-3	2-of-3	BIG1	Fully stag.	Pair-stag.
Normal vs. target only	0.94±0.02	0.13±0.00	0.55±0.03	0.65±0.08	0.14±0.02	0.27±0.02
+ fully staggered negatives	0.95±0.01	0.06±0.01	0.40±0.05	0.08±0.04	0.02±0.01	0.12±0.05
+ generic negatives (OE-style)	0.93±0.01	0.13±0.00	0.56±0.01	0.70±0.03	0.13±0.01	0.26±0.04
+ channel near-misses	0.83±0.06	0.04±0.02	0.15±0.05	0.01±0.02	0.04±0.02	0.08±0.04
+ CIS (channel + pair-staggered)	0.90±0.00	0.05±0.01	0.23±0.03	0.04±0.03	0.03±0.02	0.04±0.02
Reference all-sensors rule	0.89±0.03	0.05±0.01	0.21±0.04	0.05±0.02	0.09±0.03	0.33±0.06

the measured $R/R_{\min} \in [1.19, 1.34]$. The near-misses are flagged in 98–100% of draws. After retraining with 100 near-misses per type, path norms are 4.0–8.6 times the minimum, above the requirement $k\gamma/G$ of theorem 5(b) in every run. Even so, the fraction of near-misses still flagged grows from 0.20 at $k = 2$ to 1.00 at $k = 8$. At a fixed training budget, rejecting more near-miss types is harder, which is consistent with the growing norm cost, although this experiment does not isolate the cause.

Which negatives teach what (theorem 6). In the synthetic temporal experiment (fig. 1b), fully staggered negatives remove fully staggered false alarms but leave pair-staggered look-alikes high. Pair-staggered negatives remove both, as theorem 6(b,c) predicts. The same ordering holds on all three real FCU runs: pair-staggered look-alikes are flagged at 0.37–0.59 after fully staggered negatives vs. 0.00–0.06 after CIS, with detection at 1.00. Fully staggered negatives do lower pair-staggered alarms partly (from 0.92–0.99 on the FCU). Theorem 6(c) does not exclude this: it says these labels do not identify simultaneity, not that a learner cannot generalize from them. In the toy replica (table 1), fully staggered negatives remove fully staggered alarms ($0.14 \to 0.02$). Because each of their segments shifts a single sensor, they also remove most single-sensor alarms (BIG1 $0.65 \to 0.08$), but they leave 2-of-3 at 0.40. Channel near-misses bring 2-of-3 to 0.15 at a detection cost ($0.94 \to 0.83$). CIS reaches 2-of-3 0.23 and pair-staggered 0.04 at detection 0.90. This is close to the reference rule on 2-of-3 (0.21) and better on pair-staggered windows (0.33). Generic negatives on unrelated sensors leave every rate unchanged (2-of-3 0.56). They carry no information about the conjunction, unlike outlier exposure in standard OOD detection. Table 8 shows the same ordering for all seven toy models.

How many examples (proposition 7). Near-miss sweeps on real data (fig. 1c) show that about 300 examples per type bring FCU 2-of-3 alarms from 0.96–1.00 to 0.06–0.14. On the AHU coil-valve target, whose third signature sensor moves only 0.37 s.d., 1000 per type still leave 0.50. Proposition 7 predicts this weak-sensor bottleneck: relative to a sensor moving one s.d., it implies roughly $(1/0.37)^2 \approx 7$ times more examples. In the synthetic sweep (table 6), stronger weight decay slows the decrease (0.46 vs. 0.34 at 256 per type), in the direction of proposition 12. On real data the ordering reverses: the LSTM with weight decay 10^{-3} needs fewer near-misses than without (FCU fan, clean: 0.06 vs. 0.44 at 300 per type). Proposition 12 concerns the regularized optimum of a two-layer net and does not describe this case.

Real never-seen faults. On the FCU, faults that share signature sensors are flagged far more often than faults that share none (table 2). Most fleet false alarms (72–85%) then come from other faults, not from normal operation. On the AHU, other faults rarely move the sensors that the LSTM relies on, and false alarms on never-seen faults stay $\leq 6\%$. The law predicts exactly this: what matters is the overlap r that real look-alikes have with the evidence the model sums. Adding 2-of-3 near-misses to FCU fan-outlet training also cuts flag rates on never-seen faults sharing one or two sensors, from $0.63/0.79$ to $0.11/0.22$, although none of these faults was used in training. Across $n \in \{500, 2000, 8000\}$ and prevalences $\{0.1, 0.5, 0.9\}$ the LSTM's 2-of-3 rate never fell below 0.65 on any target, as theorem 1(ii–iii) predicts for more data and rebalancing.

Limitations. The real data are simulations with less sensor noise than field data. The signature and the reference rule are estimates, and real-data runs use one seed. Theorems 1, 2 and 4 assume

Table 2: Real faults (LBNL simulations, LSTM, one seed). *2-of-3*: look-alike flag rate after normal-vs-target training (reference all-sensors rule in brackets) and after adding channel near-misses (+NM). *Pair-stag.*: after fully staggered negatives vs. after CIS. *Never-seen*: other fault files sharing one or two signature sensors vs. none. *Fleet share*: fraction of false alarms caused by other faults in an 89/1/10% normal/target/other mix.

		2-of-3 flagged		Pair-stag. flagged	Never-seen faults	Fleet
Target (signature shifts, s.d.)	Det.	plain (ref.)	+ NM	stag. / CIS	partial / none	share
FCU fan outlet, clean (2.6, 2.3, 1.7)	1.00	0.97 (0.35)	0.01	0.51 / 0.00	0.70 / 0.22	76%
FCU fan outlet, real-mode	1.00	1.00 (0.30)	0.01	0.37 / 0.00	0.67 / 0.25	72%
FCU cooling valve (9.0, 8.1, 2.3)	1.00	1.00 (0.43)	0.23	0.59 / 0.06	0.60 / 0.18	85%
AHU coil valve (1.8, 0.8, 0.37)	1.00	0.78 (0.74)	0.30	0.18 / 0.15†	0.03 / 0.00	33%
AHU damper (2.1, 0.9, 0.6)	1.00	0.79	0.19	–	0.06 / 0.01	–

†Even the reference rule flags 0.90 of AHU coil PAIRSTAG windows, so the timing test is uninformative there. "–": not run with the new constructs.

Gaussian noise and a known onset. Theorem 5 is a bound in terms of a measured constant, not a convergence theorem for gradient descent. Theorem 6 assumes per-axis monotone learners. Where real detection saturates, the law cannot be tested quantitatively. The weight-decay prediction of proposition 12 did not transfer to the real-data LSTM.

7 Conclusion

Detectors trained on normal-vs-target data learn to add up the right evidence instead of requiring all of it. For the training question this is optimal. For deployment it is wrong, and more data cannot change it. The theory gives three practical consequences. A model's look-alike false alarms can be predicted from its detection rate. Uncapped additive and channel-independent designs cannot be made safe against loud look-alikes. And the fix is small and specific: k channel near-misses and $\binom{k}{2}$ pair-staggered near-misses, with the weakest sensor setting the amount. Open problems include a gradient-descent convergence theorem for non-symmetric data, the unknown-onset version of the law, and training procedures that make capped models learn the conjunction they can represent.

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A PROOFS

Throughout, Φ and φ are the standard normal CDF and density, $z_\alpha = \Phi^{-1}(1-\alpha)$, and $\|\cdot\|$ is the Euclidean norm. We use two elementary facts.

(F1) If $x \sim \mathcal{N}(\mu, \Sigma)$ and $u \in \mathbb{R}^d$, then $u^\top x \sim \mathcal{N}(u^\top \mu,\, u^\top \Sigma u)$.

(F2) (Neyman–Pearson lemma, Neyman & Pearson, 1933; Lehmann & Romano, 2005.) For two distributions with densities p_0, p_1, a test that rejects when p_1/p_0 exceeds a constant chosen to give size α is most powerful among all tests of size $\le \alpha$. Ties occur with probability 0 when the likelihood ratio has a continuous distribution.

A.1 PROOF OF THEOREM 1

(i) Under Assumption 1 the densities are $p_h(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-\frac{1}{2}(x-h)^\top\Sigma^{-1}(x-h))$. Hence

$$\log\frac{p_g(x)}{p_0(x)} = -\tfrac{1}{2}(x-g)^\top\Sigma^{-1}(x-g) + \tfrac{1}{2}x^\top\Sigma^{-1}x = g^\top\Sigma^{-1}x - \tfrac{1}{2}g^\top\Sigma^{-1}g.$$

By Bayes' rule the log posterior odds equal the log-likelihood ratio plus $\log\frac{\pi}{1-\pi}$, which gives ℓ_π. If $\Sigma = \mathrm{blockdiag}(\Sigma_{11}, \ldots, \Sigma_{CC})$ over channels, then $g^\top\Sigma^{-1}x = \sum_j g_j^\top\Sigma_{jj}^{-1}x_j$, where $x_j, g_j \in \mathbb{R}^T$ are the channel-j blocks. This is a sum of per-channel terms, and $g_j = 0$ for $j \notin S$.

(ii) By (i) the likelihood ratio is a strictly increasing function of $u(x) = g^\top\Sigma^{-1}x$. By (F2), the most powerful size-α test rejects when $u(x) > t$, with t chosen so that $P_0(u > t) = \alpha$. By (F1), under P_0, $u \sim \mathcal{N}(0, g^\top\Sigma^{-1}\Sigma\Sigma^{-1}g) = \mathcal{N}(0, D^2)$, so $t = z_\alpha D$. Neither the test nor its threshold involves π.

(iii) Let $P_\pi = (1-\pi)P_0 + \pi P_g$ and let $\hat{t}_n$ be the $(1-\alpha)$-quantile of $\hat{f}_n$ under P_0. Let t^* be the $(1-\alpha)$-quantile of ℓ_π under P_0. Since ℓ_π is an increasing affine function of u, $\{\ell_\pi > t^*\} = \{\phi^* = 1\}$ by (ii).

Step 1 (convergence under P_0). $P_0(A) \le P_\pi(A)/(1-\pi)$ for every event A, so $\hat{f}_n \to \ell_\pi$ in P_π-probability implies convergence in P_0-probability.

Step 2 (threshold convergence). Let F_n and F be the CDFs of $\hat{f}_n$ and ℓ_π under P_0. Convergence in probability implies convergence in distribution, so $F_n(t) \to F(t)$ at every t, because F is continuous: ℓ_π is Gaussian under P_0 with positive variance D^2. F is also strictly increasing. Fix $\epsilon > 0$. Then $F(t^*-\epsilon) < 1-\alpha < F(t^*+\epsilon)$, and for n large $F_n(t^*-\epsilon) < 1-\alpha < F_n(t^*+\epsilon)$. Hence $\hat{t}_n \in [t^*-\epsilon, t^*+\epsilon]$, where $\hat{t}_n = \inf\{t : F_n(t) \ge 1-\alpha\}$. So $\hat{t}_n \to t^*$. If $\hat{t}_n$ is instead the empirical quantile of $m_n \to \infty$ normal scores drawn independently of training, then conditionally on $\hat{f}_n$ these scores are i.i.d. with CDF F_n, and the Dvoretzky–Kiefer–Wolfowitz inequality (Massart, 1990) gives $\Pr(\sup_t|\hat{F}_{m_n} - F_n| > \epsilon') \le 2e^{-2m_n\epsilon'^2}$ uniformly in $\hat{f}_n$. Taking ϵ' below the two gaps $|F_n(t^* \pm \epsilon) - (1-\alpha)|$, which stay bounded away from 0 for large n, the same sandwich holds for $\hat{F}_{m_n}$ with probability $\to 1$, so $\hat{t}_n \to t^*$ in probability.

Step 3 (transfer to Q). P_π has an everywhere positive density, so Lebesgue measure is absolutely continuous w.r.t. P_π. Hence $Q \ll P_\pi$. For finite measures this means: for every $\eta > 0$ there is $\gamma > 0$ with $P_\pi(A) < \gamma \Rightarrow Q(A) < \eta$. Therefore $\hat{f}_n \to \ell_\pi$ in Q-probability.

Step 4 (disagreement). For any $\epsilon > 0$, the event $\{\hat{\phi}_n \ne \phi^*\}$ is contained in

$$\{|\ell_\pi - t^*| \le 2\epsilon\} \cup \{|\hat{f}_n - \ell_\pi| > \epsilon\} \cup \{|\hat{t}_n - t^*| > \epsilon\}.$$

To see this, suppose $|\ell_\pi - t^*| > 2\epsilon$, $|\hat{f}_n - \ell_\pi| \le \epsilon$ and $|\hat{t}_n - t^*| \le \epsilon$. Then $\hat{f}_n - \hat{t}_n$ and $\ell_\pi - t^*$ have the same sign, so the two tests agree. The third event has probability $\to 0$ (Step 2; it is eventually empty for the population quantile). The second has Q-probability $\to 0$ (Step 3). The first has Q-probability $\to 0$ as $\epsilon \to 0$, because ℓ_π is a non-constant affine function of x and so its level sets have Lebesgue measure zero, hence Q-measure zero. Taking $n \to \infty$ and then $\epsilon \to 0$ gives $Q(\hat{\phi}_n \ne \phi^*) \to 0$. □

Remark 8. Logistic regression on $(x, 1)$ is well specified by (i), and its maximum-likelihood estimator is consistent, so it satisfies the premise of (iii). The same conclusion holds for exponential-family

noise whose sufficient statistics are channel-separable: the log-likelihood ratio is then a sum of per-channel terms.

A.2 Proof of theorem 2 and corollary 3

Under $\mathcal{N}(h, \Sigma)$, (F1) gives $u = g^\top \Sigma^{-1} x \sim \mathcal{N}(g^\top \Sigma^{-1} h,\, D^2) = \mathcal{N}(D^2 r(h), D^2)$. Thus $\Pr_h(u > z_\alpha D) = \Pr(Z > z_\alpha - Dr(h)) = \Phi(Dr(h) - z_\alpha)$ for $Z \sim \mathcal{N}(0, 1)$.

Special cases. Assume independent channels with equal per-channel detectability $D_j^2 = D^2/k$ for $j \in S$. For $h = g \odot \mathbf{1}_A$, block-diagonality gives $g^\top \Sigma^{-1} h = \sum_{j \in A} D_j^2 = |A| D^2/k$, so $r = m/k$ when $|A| = m$. Shifting only channel c, by c' times its target shift ($h = c' g_c$), gives $r = c' D_c^2 / D^2 = c'/k$. For $h = -g$, $r = -1$. For h supported on S^c, $g^\top \Sigma^{-1} h = 0$ because g vanishes off S and Σ^{-1} is block-diagonal, so the rate is $\Phi(-z_\alpha) = \alpha$.

Corollary 3. $\beta = \Phi(D - z_\alpha)$ gives $D = \Phi^{-1}(\beta) + z_\alpha$. Substituting this into the law gives the stated formula. For $r \in (0, 1)$ the argument $r(\Phi^{-1}(\beta) + z_\alpha) - z_\alpha$ is strictly increasing in β and tends to $+\infty$ as $\beta \to 1$. Φ is strictly increasing. Replacing (g, h, Σ) by (cg, ch, Σ) with $c > 1$, or by $(g, h, c\Sigma)$ with $c < 1$, leaves $r(h) = g^\top \Sigma^{-1} h / g^\top \Sigma^{-1} g$ unchanged and multiplies D by c or $c^{-1/2}$, so β and every look-alike rate with $r(h) \in (0, 1)$ increase. □

A.3 Shift statistic and proof of theorem 4

Lemma 9 (Shift statistic)**.** *Let $B, I \subset [T]$ be disjoint baseline and post-onset index sets and $c = \mathbf{1}_I/|I| - \mathbf{1}_B/|B| \in \mathbb{R}^T$. Let $a_j = s_j\, c^\top x_j$. Under Assumption 1 with block-diagonal Σ and a pattern h whose channel-j block equals $s_j \mu_j \mathbf{1}_I$ (and 0 on B), the vector $a = (a_j)_{j \in S}$ is $\mathcal{N}(\mu, \mathrm{diag}(\sigma_j^2))$ with $\sigma_j^2 = c^\top \Sigma_{jj} c$.*

Proof. $a_j = s_j c^\top x_j$ is linear in x_j, with mean $s_j c^\top (s_j \mu_j \mathbf{1}_I) = \mu_j$ because $c^\top \mathbf{1}_I = 1$ and $s_j^2 = 1$. Its variance is $c^\top \Sigma_{jj} c$ (F1). The a_j are jointly Gaussian and depend on different blocks, so block-diagonality makes them independent. □

(a) Fix $j \in S$ and let $\mu^{(j)} = \mu - \mu_j e_j$, i.e. the near-miss without sensor j, with $\mu = (\delta_i)_{i \in S}$. Since $M_i \geq \delta_i$, $\mathcal{N}(\mu^{(j)}, \cdot)$ belongs to $\mathcal{N}(M)$. A deployment-valid ϕ is therefore in particular a level-α test of $\mathcal{N}(\mu^{(j)}, \Lambda)$ against $\mathcal{N}(\mu, \Lambda)$, with $\Lambda = \mathrm{diag}(\sigma_i^2)$. These two laws differ only in the mean of coordinate j. By (i) of theorem 1 applied to $g = \mu - \mu^{(j)} = \delta_j e_j$, the log-likelihood ratio is $\delta_j (a_j - \mu_j^{(j)})/\sigma_j^2$ plus a constant. By (F2) the most powerful level-α test rejects when $a_j > z_\alpha \sigma_j$, with power $\Phi(\delta_j/\sigma_j - z_\alpha)$. So no deployment-valid test exceeds this power. Minimize over j. *Full windows:* the same argument applies to any measurable function of the whole window x. The target pattern g and the near-miss pattern $g - g_j$ differ only in the channel-j block, so with block-diagonal Σ the likelihood ratio depends on x_j only, and the most powerful test has power $\Phi(D_j - z_\alpha)$ with $D_j^2 = g_j^\top \Sigma_{jj}^{-1} g_j$.

(b) *Validity.* Let $Q \in \mathcal{N}(\infty)$ have support $A \subsetneq S$, and pick $j \in S \setminus A$, so $\mu_j = 0$ under Q. Then $Q(\phi_{IU} = 1) \leq Q(a_j/\sigma_j > z_\alpha) = \alpha$. Under normal ($\mu = 0$) the same bound holds. *Power.* Under the target the a_j are independent with $\Pr(a_j/\sigma_j > z_\alpha) = \Phi(D_j - z_\alpha)$, so the power is $\prod_j \Phi(D_j - z_\alpha)$. *Approximation.* Write $q_j = \Phi(z_\alpha - D_j)$ for the per-coordinate miss. The IU miss is $1 - \prod_j (1 - q_j) \leq \sum_j q_j \leq k \max_j q_j$, using the union bound. By (a), every valid detector misses with probability at least $1 - \min_j \Phi(D_j - z_\alpha) = \max_j q_j$.

(c) Let $w \geq 0$, $w \neq 0$, with equal δ, σ and $M_i = \delta$. For each i, the near-miss $\mu^{(i)} = \delta(\mathbf{1} - e_i)$ is in $\mathcal{N}(\delta)$, and $w^\top a \sim \mathcal{N}(\delta(\|w\|_1 - w_i), \sigma^2 \|w\|^2)$ under it. Validity requires $\Pr(w^\top a > t) \leq \alpha$, i.e. $t \geq \delta(\|w\|_1 - w_i) + z_\alpha \sigma \|w\|$ for all i. The binding constraint is $i^* = \arg\min_i w_i$. The power is $\Phi\big((\delta\|w\|_1 - t)/(\sigma\|w\|)\big) \leq \Phi\big(\delta \min_i w_i/(\sigma\|w\|) - z_\alpha\big)$. Finally, $k \min_i w_i^2 \leq \|w\|^2$ gives $\min_i w_i \leq \|w\|/\sqrt{k}$, with equality iff all w_i are equal. Other members of $\mathcal{N}(\delta)$ have coordinatewise smaller means, hence smaller false-alarm rates (since $w \geq 0$), so they do not tighten the constraint.

(d) Let ψ_j be nondecreasing, real-valued and unbounded above. Since $k \geq 2$, the vector Ae_j (all sensors normal except j, shifted by $A > 0$) has support $\{j\} \subsetneq S$, so it belongs to $\mathcal{N}(\infty)$ for every A. Fix any $t \in \mathbb{R}$ and $\epsilon > 0$. The random variable $Y = \sum_{i \neq j} \psi_i(a_i)$ has a fixed distribution under these alternatives and is a.s. finite, so there is y_ϵ with $\Pr(Y < -y_\epsilon) < \epsilon$. Since ψ_j is unbounded and nondecreasing, there is u_ϵ with $\psi_j(u) > t + y_\epsilon$ for all $u \geq u_\epsilon$. Under Ae_j, $\Pr(a_j \geq u_\epsilon) \to 1$ as $A \to \infty$. Hence $\liminf_{A\to\infty} \Pr(\sum_i \psi_i(a_i) > t) \geq 1 - \epsilon$. So no finite threshold is valid, and the only valid choice $t = +\infty$ (never alarm) has power 0. Linear detectors with $w_j > 0$ are the case $\psi_j(u) = w_j u$. For a channel-independent model $f(X) = \sum_j f_j(X_{\cdot j})$, apply the same argument with the null member that shifts channel j by $A\, s_j \mathbf{1}_I$. "Unbounded along fault-direction shifts" means $f_j(X_{\cdot j} + A s_j \mathbf{1}_I) \to \infty$ in probability as $A \to \infty$.

(e) With $\psi_j = \mathbf{1}\{a_j > z_\alpha \sigma_j\}$ and $t = k - \frac{1}{2}$, $\sum_j \psi_j > t$ iff all indicators equal 1, i.e. $\phi_{IU} = 1$.

(f) Apply theorem 2 with $r = (k-1)/k$ and $D = \sqrt{k}\,\delta/\sigma$: $Dr = (k-1)\delta/(\sigma\sqrt{k}) > 0$, so the rate exceeds $\Phi(-z_\alpha) = \alpha$. □

A.4 Norms, Lipschitz constants, and proof of theorem 5

Lemma 10 (Norms bound the Lipschitz constant)**.** *For $f(x) = \sum_i v_i \sigma(w_i^\top x + b_i) + c$ with σ 1-Lipschitz, $|f(x) - f(y)| \leq R\|x - y\|$ with $R = \sum_i |v_i| \|w_i\|$. Moreover,* min *over the per-unit rescalings $(v_i, w_i, b_i) \mapsto (v_i/s, s w_i, s b_i)$, $s > 0$, of $\frac{1}{2}\sum_i (v_i^2 + \|w_i\|^2 + b_i^2)$ is at least R. For ReLU these rescalings leave f unchanged, so ℓ_2 weight decay upper-bounds λR at any minimizer. For depth-L networks,* $\mathrm{Lip}(f) \leq \prod_l \|W_l\|_{\mathrm{op}} \leq (\|\theta\|^2/L)^{L/2}$.

Proof. $|f(x) - f(y)| \leq \sum_i |v_i| |\sigma(w_i^\top x + b_i) - \sigma(w_i^\top y + b_i)| \leq \sum_i |v_i| |w_i^\top (x - y)| \leq R\|x - y\|$ by Cauchy–Schwarz. For one unit, $\min_{s>0} \frac{1}{2}(v^2/s^2 + s^2(\|w\|^2 + b^2)) = |v|\sqrt{\|w\|^2 + b^2} \geq |v|\|w\|$, by AM–GM with equality at $s^2 = |v|/\sqrt{\|w\|^2 + b^2}$. ReLU is positively homogeneous, so the rescaling preserves f. For depth L, the Lipschitz constant of a composition is at most the product of the factors' constants. Then $\|W_l\|_{\mathrm{op}} \leq \|W_l\|_F$, and AM–GM gives $\prod_l \|W_l\|_F \leq (\sum_l \|W_l\|_F^2/L)^{L/2}$. □

Theorem 5(a). By definition of the Lipschitz constant, $f(x) \geq f(x_+) - \mathrm{Lip}(f)\|x - x_+\| = f(x_+) - \kappa \frac{G}{\Delta} \rho \Delta = f(x_+) - \kappa\rho G$. Subtracting $f(x_0)$ and dividing by G gives $p(x) \geq 1 - \kappa\rho$. *Distances.* Take $x_0 = 0$ (baseline-corrected normal) and $x_+ = g$, with every signature channel shifted by δ on a block $\mathcal{B}$ of L steps. Then $\Delta = \delta\sqrt{kL}$. The channel-j near-miss $x^{(j)} = g - g_j$ satisfies $\|x^{(j)} - x_+\| = \|g_j\| = \delta\sqrt{L}$, so $\rho = 1/\sqrt{k}$. Displacing channel j's block by $\Delta t \leq L$ steps changes Δt entries from δ to 0 and Δt entries from 0 to δ, so the distance is $\delta\sqrt{2\Delta t}$ and $\rho = \sqrt{2\Delta t/(kL)}$. *Interpretation of κ.* Every f with $f(x_+) - f(x_0) = G$ has $\mathrm{Lip}(f) \geq G/\Delta$, and the linear score $f(x) = \frac{G}{\Delta^2}(x_+ - x_0)^\top (x - x_0)$ attains it. So $\kappa \geq 1$ measures excess slope. In experiments we upper-bound κ by $R/R_{\min}$ using lemma 10.

Theorem 5(b). For each unit and each j, $|\mathrm{relu}(w_i^\top x_+ + b_i) - \mathrm{relu}(w_i^\top x^{(j)} + b_i)| \leq |w_i^\top (x_+ - x^{(j)})| = \delta |\sum_{t \in \mathcal{B}} w_{i,tj}| \leq \delta \sum_{t \in \mathcal{B}} |w_{i,tj}|$, where $w_{i,tj}$ is the weight of unit i on input (t, j). Summing over units with weights $|v_i|$ gives $\gamma \leq f(x_+) - f(x^{(j)}) \leq \delta \sum_i |v_i| \sum_{t \in \mathcal{B}} |w_{i,tj}|$. Summing over $j \in S$,

$$k\gamma \leq \delta \sum_i |v_i|\, \|w_{i,\mathcal{B}\times S}\|_1 \leq \delta\sqrt{kL} \sum_i |v_i|\, \|w_{i,\mathcal{B}\times S}\|_2 \leq \delta\sqrt{kL}\, R,$$

using Cauchy–Schwarz on the kL coordinates of $\mathcal{B} \times S$. Hence $R \geq \sqrt{k}\gamma/(\delta\sqrt{L})$. With $R_{\min} = G/\Delta = G/(\delta\sqrt{kL})$ this is $R \geq k\frac{\gamma}{G} R_{\min}$. For unequal shifts the same steps give $R \geq \gamma \sum_{j \in S} \delta_j^{-1}/\sqrt{kL}$. □

Remark 11 (Exact linear limit). Lyu et al. (2021) show that, for linearly separable data that is symmetric (closed under $(x, y) \mapsto (-x, -y)$), gradient flow from small initialization on two-layer leaky-ReLU networks converges in direction to the linear max-margin classifier. Symmetry holds exactly after centring at the midpoint of the prototypes and symmetrizing the sample. For two prototypes $\pm(x_+ - x_0)/2$ that classifier is proportional to $x_+ - x_0 \propto \mathbf{1}_S$ (in shift-statistic space), so channel

near-misses sit at $p = (k-1)/k$. The midpoint threshold is $p = \frac{1}{2}$, which they exceed for every $k \geq 3$. We do not rely on this result for non-symmetric data. Theorem 5 holds for any network, in terms of its measured κ.

Proposition 12 (Necessary near-miss fraction under weight decay)**.** *In shift-statistic space, take noise-free prototypes with mass q at $x_0 = 0$ (label -1), q at $x_+ = \delta \mathbf{1}_S$ (label $+1$), and p/k at each channel near-miss (label -1), with $2q + p = 1$. Let $\ell(z) = \log(1 + e^{-z})$ and consider two-layer ReLU networks with objective $J(\theta) = \sum \text{mass} \cdot \ell(y f_\theta(x)) + \lambda R(\theta)$. Define*

$$J_{\text{cheap}}(p) = \min_{m_0, m_1 \geq 0} \; q\ell(m_0) + q\ell(m_1) + p\,\ell\big(-m_1 + \tfrac{m_0+m_1}{k}\big) + \lambda \tfrac{m_0+m_1}{\delta\sqrt{k}},$$

$$B(p) = \inf_{m, m_0 \in \mathbb{R}} \; q\ell(m) + q\ell(m_0) + \lambda \max\Big(\tfrac{\sqrt{k}\, m_+}{\delta}, \tfrac{|m+m_0|}{\delta\sqrt{k}}\Big).$$

If $J_{\text{cheap}}(p) < B(p)$, every minimizer of J flags at least one near-miss ($f(x^{(j)}) > 0$). The same holds with λR replaced by ℓ_2 weight decay $\frac{\lambda}{2}\sum_i (v_i^2 + \|w_i\|^2 + b_i^2)$ with an unpenalized output bias: by lemma 10 the weight decay of every θ is at least $\lambda R(\theta)$, and for the one-unit network of the upper bound, which has hidden bias 0, the best rescaling makes it equal to λR.

Proof. Upper bound. The one-unit network $f(x) = v\,\text{relu}(\mathbf{1}_S^\top x/\sqrt{k}) - m_0$ with $v = (m_0 + m_1)/(\delta\sqrt{k})$ has $f(x_0) = -m_0$, $f(x_+) = m_1$ and $f(x^{(j)}) = v\delta(k-1)/\sqrt{k} - m_0 = m_1 - (m_0 + m_1)/k$. Its path norm is v, since $\|\mathbf{1}_S/\sqrt{k}\| = 1$. So $\min J \leq J_{\text{cheap}}(p)$. *Lower bound.* Let f reject every near-miss, $f(x^{(j)}) \leq 0$, and write $m = f(x_+)$, $m_0 = -f(x_0)$. The near-miss losses are nonnegative, so they can be dropped. Theorem 5(b) with $\gamma = m_+$ ($L = 1$) gives $R \geq \sqrt{k}\, m_+/\delta$. Lemma 10 with $\|x_+ - x_0\| = \delta\sqrt{k}$ gives $R \geq |m + m_0|/(\delta\sqrt{k})$. Hence $J(\theta) \geq B(p)$, and when $J_{\text{cheap}}(p) < B(p)$ no near-miss-rejecting network attains $\min J$. □

Numerically ($\delta = 1$), the largest p with $J_{\text{cheap}} < B$ is $p^* = 2 \times 10^{-4}, 1.3 \times 10^{-3}, 8.8 \times 10^{-3}, 1.7 \times 10^{-2}$ for $\lambda = 10^{-4}, 10^{-3}, 10^{-2}, 3 \times 10^{-2}$ at $k = 3$, and p^* grows with k (to 6.1×10^{-2} at $k = 8$, $\lambda = 3 \times 10^{-2}$). This is a *necessary* condition only. With noisy data, gradient-trained networks needed far more near-misses (table 6).

A.5 Proof of theorem 6

Setting. The labelled examples are configurations labelled by the target rule h^*. A labelled set $\mathcal{L}$ identifies h^* within $\mathcal{H}$ if h^* is the only $h \in \mathcal{H}$ with $h(c) = h^*(c)$ for all $c \in \mathcal{L}$. *Channel axis.* Configurations are $A \subseteq S$ (sensors shifted, simultaneously). $\mathcal{H}_{\text{ch}}$ is the set of $h : 2^S \to \{0, 1\}$ that are monotone ($A \subseteq A' \Rightarrow h(A) \leq h(A')$) with $h(S) = 1$. The target is $h^*_{\text{ch}}(A) = \mathbf{1}\{A = S\}$, and training data label $A = \emptyset$ (normal) as 0. *Time axis.* All of S is shifted on nonempty half-open intervals $I_j = [l_j, u_j)$. Configurations are the realizable overlap graphs $\mathcal{G}$ on vertex set S. $\mathcal{H}_{\text{t}}$ is the set of h monotone in the edge set with $h(K_k) = 1$. The target "some moment at which all k are shifted" is h^*_{t}.

(a) *Sufficiency.* Let $\mathcal{L} \ni \emptyset$ contain every $S \setminus \{j\}$. Any $A \neq S$ omits some j, so $A \subseteq S \setminus \{j\}$, and monotonicity gives $h(A) \leq h(S \setminus \{j\}) = 0$ for every consistent h. Together with $h(S) = 1$ this forces $h = h^*_{\text{ch}}$. *Necessity.* If $S \setminus \{i\} \notin \mathcal{L}$, let $h_i(A) = \mathbf{1}\{A \supseteq S \setminus \{i\}\}$. It is monotone, $h_i(S) = 1$, and $h_i(\emptyset) = 0$ since $k \geq 2$. It differs from h^*_{ch} exactly at $A = S \setminus \{i\}$, so it agrees with h^*_{ch} on $\mathcal{L}$ and $\mathcal{L}$ does not identify. (This characterization of teaching sets for monotone rules is classical, cf. Goldman & Kearns, 1995.)

(b) *Helly on the line.* If $I_i \cap I_j \neq \emptyset$ for all i, j, then $l_i < u_j$ for all i, j, so $\max_i l_i < \min_j u_j$, and every point of $[\max_i l_i, \min_j u_j)$ lies in all intervals. Conversely a common point lies in every pairwise intersection. Hence $h^*_{\text{t}}(\mathcal{G}) = \mathbf{1}\{\mathcal{G} = K_k\}$, the conjunction of the $\binom{k}{2}$ edge indicators. *Sufficiency.* If $\mathcal{L}$ contains $K_k - e$ for every edge e, any realizable $\mathcal{G} \neq K_k$ misses some edge e, so $\mathcal{G} \subseteq K_k - e$ and $h(\mathcal{G}) \leq h(K_k - e) = 0$. *Necessity.* If $K_k - e \notin \mathcal{L}$, let $h_e(\mathcal{G}) = \mathbf{1}\{\mathcal{G} \supseteq K_k - e\}$. It is monotone with $h_e(K_k) = 1$ and differs from h^*_{t} only at $K_k - e$, so it is consistent with $\mathcal{L}$. *Realizability with equal durations.* For $e = \{a, b\}$ take $I_a = [0, 1)$, $I_b = [1, 2)$ and $I_j = [\frac{1}{2}, \frac{3}{2})$ otherwise. Then $I_a \cap I_j = [\frac{1}{2}, 1)$, $I_b \cap I_j = [1, \frac{3}{2})$, $I_j \cap I_{j'} = I_j$ and $I_a \cap I_b = \emptyset$, so $\mathcal{G} = K_k - e$. (Algorithm 1 uses exactly this construction scaled to the post-onset window.)

(c) The fully staggered configuration has $\mathcal{G} = \emptyset$, the minimum of the edge order. Labelling it 0 excludes exactly the rules with $h(\emptyset) = 1$. By monotonicity that is only the constant rule $h \equiv 1$. For $k \geq 3$, $K_k - e \neq \emptyset$, so every witness h_e in the necessity proof of (b) satisfies $h_e(\emptyset) = 0$ and survives. Fully staggered negatives therefore neither identify the rule nor replace any pair-staggered pattern. (For $k = 2$, $K_2 - e = \emptyset$ and the two notions coincide.)

(d) Let $\mathcal{C}_{\mathrm{ch}}$ and $\mathcal{C}_{\mathrm{t}}$ be the two configuration sets, sharing only the target (all k sensors shifted and overlapping). Let $\mathcal{H}$ contain the rules on $\mathcal{C}_{\mathrm{ch}} \cup \mathcal{C}_{\mathrm{t}}$ whose restrictions lie in $\mathcal{H}_{\mathrm{ch}}$ and $\mathcal{H}_{\mathrm{t}}$. The constraints decouple, since every member equals 1 at the shared target. So $\mathcal{L}$ identifies h^* in $\mathcal{H}$ iff $\mathcal{L} \cap \mathcal{C}_{\mathrm{ch}}$ and $\mathcal{L} \cap \mathcal{C}_{\mathrm{t}}$ identify on their axes. By (a) and (b) this needs and is achieved by the k channel and the $\binom{k}{2}$ pair-staggered near-misses. □

Remark 13 (Joint monotonicity). Suppose instead that rules are monotone in the joint order $(A, \mathcal{G}) \preceq (A', \mathcal{G}')$ iff $A \subseteq A'$ and $\mathcal{G} \subseteq \mathcal{G}'$, over all configurations. Then the channel near-miss $(S \setminus \{j\}, K_{S\setminus\{j\}})$ lies below $(S, K_k - \{i, j\})$ for any $i \neq j$. So the $\binom{k}{2}$ pair-staggered negatives would already force every channel near-miss to 0. We do not assume joint monotonicity for learned networks, and we did not test pair-staggered negatives alone. Our real-data remedy (CIS) supplies both types.

A.6 Proof of proposition 7

Let $\bar{x}_j$ and $\bar{y}_j$ be the sample means of sensor j over n i.i.d. near-misses of type j and n targets, with true means ν_j and $\nu_j + \Delta_j$ and noise $\mathcal{N}(0, \sigma_j^2)$. The threshold is $\hat{c}_j = (\bar{x}_j + \bar{y}_j)/2$, and the midpoint is $c_j = \nu_j + \Delta_j/2$. Then $|\hat{c}_j - c_j| \leq \frac{1}{2}(|\bar{x}_j - \nu_j| + |\bar{y}_j - \nu_j - \Delta_j|) \leq \max(|\bar{x}_j - \nu_j|, |\bar{y}_j - \nu_j - \Delta_j|)$. The Gaussian tail bound gives $\Pr(|\bar{x}_j - \nu_j| > t) \leq 2e^{-nt^2/(2\sigma_j^2)}$, and similarly for $\bar{y}_j$. With $t = \Delta_j/4$, $\Pr(|\hat{c}_j - c_j| > \Delta_j/4) \leq 4e^{-n\Delta_j^2/(32\sigma_j^2)}$. A union bound over $j \in S$ shows that all thresholds are within $\Delta_j/4$ of their midpoints with probability $\geq 1 - 4k\, e^{-n \min_j \Delta_j^2/(32 \max_j \sigma_j^2)} \geq 1 - \eta$ under the stated condition on n. On this event $\hat{c}_j \geq \nu_j + \Delta_j/4$. A new near-miss of type j is flagged only if its sensor-j value exceeds $\hat{c}_j$, which has probability $\leq \Phi(-\Delta_j/(4\sigma_j))$. Symmetrically $\hat{c}_j \leq \nu_j + \frac{3}{4}\Delta_j$, so the target fails sensor j with probability $\leq \Phi(-\Delta_j/(4\sigma_j))$. A union bound over j bounds the miss probability. If the n values are consecutive draws of a stationary AR(1) sequence with $\phi \in [0, 1)$, the variance of their mean is $\frac{\sigma_j^2}{n}\big(1 + 2\sum_{h=1}^{n-1}(1 - \frac{h}{n})\phi^h\big) \leq \frac{\sigma_j^2}{n}\frac{1+\phi}{1-\phi}$ for every n (lemma 15). The mean is still Gaussian, since it is a linear functional, so the tail bound holds with n replaced by $n(1-\phi)/(1+\phi)$. □

A.7 Window statistics and temporal dependence

Lemma 14 (Window statistics are timing- and sign-blind)**.** *Let $\psi : \mathbb{R}^T \to \mathbb{R}^q$ be invariant to permutations of the time index (e.g. mean, standard deviation, minimum, maximum, quantiles), and $\Psi(X) = (\psi(X_{\cdot 1}), \ldots, \psi(X_{\cdot C}))$. Suppose the noise has exchangeable time entries within each channel and independent channels. (i) If two patterns G, G' have, channel by channel, the same multiset of entries, then $\Psi(W + G) \overset{d}{=} \Psi(W + G')$. In particular, simultaneous, fully staggered and pair-staggered patterns with equal per-channel durations are indistinguishable by any detector built on Ψ. (ii) If $W \overset{d}{=} -W$ and $\psi(-x) = \psi(x)$ (dispersion statistics), then $\psi(W + G) \overset{d}{=} \psi(W - G)$.*

Proof. (i) For each j choose a permutation π_j with $G'_{\cdot j} = G_{\pi_j(\cdot), j}$. Exchangeability gives $W_{\cdot j} \overset{d}{=} W_{\pi_j(\cdot), j}$, and independence across channels makes this joint over j. Hence $(W + G')_{\cdot j} \overset{d}{=} (W + G)_{\pi_j(\cdot), j}$ jointly, and ψ is permutation-invariant. (ii) $\psi(W + G) = \psi(-W - G) \overset{d}{=} \psi(W - G)$. □

With AR noise, or with order-dependent statistics such as a slope or a last-m mean, (i) holds only approximately. Consistent with this, the window-statistics models flag consecutive-segment staggered faults at 0.57–1.00 on the real targets of table 9.

Lemma 15 (Temporal dependence enters through the effective noise)**.** *For stationary noise with autocovariance $\gamma(\cdot)$, the shift statistic of lemma 9 has $\sigma_j^2 = c^\top \Gamma c$ with $\Gamma_{st} = \gamma(s - t)$. For AR(1) noise with coefficient $\phi \in (-1, 1)$ and marginal variance σ_x^2, the variance of an L-step mean is*

$\sigma_x^2 L^{-2} \sum_{s,t} \phi^{|s-t|}$, *and* L *times this tends to* $\sigma_x^2(1+\phi)/(1-\phi)$ *as* $L \to \infty$. *All results above hold with these effective variances. For a post-onset mean against a known baseline* ($c = \mathbf{1}_I/|I|$) *and* $\phi \in [0, 1)$, *a larger* ϕ *raises* σ_j^2 *and lowers* D, *which moves a detector* along *the curve of corollary 3: detection and look-alike alarms both fall. This is not monotone for every statistic. Correlated noise partly cancels in the baseline-corrected difference, and for a unit step on the last 28 of* $T = 48$ *steps (unit marginal variance) the full-window* D^2 *falls from* 28 *at* $\phi = 0$ *to* 6.7 *at* $\phi = 0.9$ *and rises to* 50 *at* $\phi = 0.99$.

Proof. a_j is a linear functional of x_j (F1). For AR(1), $\gamma(h) = \sigma_x^2 \phi^{|h|}$. Then $L \cdot L^{-2} \sum_{s,t=1}^{L} \phi^{|s-t|} = 1 + 2\sum_{h=1}^{L-1}(1 - h/L)\phi^h \to 1 + 2\phi/(1-\phi) = (1+\phi)/(1-\phi)$ by dominated convergence. For $\phi \geq 0$ every term is nonnegative and at most ϕ^h, which gives the bound used in proposition 7, and each term is nondecreasing in ϕ, which gives the monotonicity claim. □

B EXPERIMENTAL DETAILS

Synthetic experiments (code: `paper_synth.py`, `theory_checks.py`). All thresholds are the 0.99 quantile of the score on 20,000 fresh normal samples, and rates are estimated on 4,000–20,000 fresh samples per construct. *Law/paradox (S1):* $d = 20$, $k = 3$, $\sigma = 1$, $\delta/\sigma \in \{0.5, \ldots, 3\}$, 2,000 + 2,000 training samples. Learners: full-batch gradient-descent logistic regression (2,000 steps, step size 0.5), and a two-layer ReLU net (64 units, initialization scale 10^{-2}, Adam, 1,500 steps, learning rate 10^{-2}). 5 seeds. *Composite (C3):* shift statistics with $k = 3$, $\sigma = 1$, 4×10^5 draws per construct; "unbounded" look-alikes use shifts of 50δ. *Separated regime (S2):* $d = 10$, $\sigma = 0.15$, $\delta = 1$, 500 + 500 samples, two-layer ReLU (64 units, initialization 10^{-3}, Adam, 3,000 steps, 10^{-2}), with 100 near-misses per type in the "with near-misses" runs. 3 seeds. *Temporal (S3):* raw windows $T = 60$, $C = 6$, $k = 3$, $\sigma = 0.5$, $\delta = 1$, shift duration $L \sim \mathcal{U}\{12, \ldots, 18\}$, random onset; 1,500 + 1,500 samples plus 600 timing negatives; two-layer ReLU on the flattened window (64 units, Adam, 1,200 steps, 3×10^{-3}, weight decay 10^{-4}). 5 seeds. *Recipe (S4):* $d = 10$, $k = 3$, $\sigma = 0.3$, 1,000 + 1,000 samples plus $c \in \{0, 4, 16, 64, 256\}$ near-misses per type, weight decay $\in \{0, 10^{-3}\}$. 3 seeds.

Toy replica (`run_toy.py`). 20 channels, 48 steps, AR(1) with $\phi = 0.7$ and unit variance, a shared slow load factor ($\phi = 0.97$, weight 0.6) on channels 0–5. The target lowers channels 0, 1, 2 by 2.5 from a random onset in $[10, 25]$. The per-window baseline (mean of the first 8 steps) is subtracted. 4,000 training windows (50/50), 2,000 normal validation windows for calibration, 2,000 test windows per construct. 3 seeds (data and initialization).

Real data (`real_data.py`, `run_real.py`). LBNL fan-coil and single-duct AHU simulations (Granderson et al., 2023), each a set of one-year runs: one fault-free run and one run per fault and severity, with identical weather and schedule. Preprocessing: resampling to 15 minutes. We drop setpoint channels, channels constant in the fault-free run, and channels that shift in the same direction in $\geq 80\%$ of fault families (a fault-free/faulty configuration mismatch in the AHU benchmark). Each channel is standardized by its fault-free statistics. *Signature*: the $k = 3$ channels with the largest mean standardized shift while the fault is visibly present, skipping near-duplicates ($|\rho| > 0.98$; disabled for the AHU coil valve, where valve position and command diverge by design). A window qualifies as a training positive only if every signature channel moves by ≥ 0.5 s.d. (0.3 for the AHU coil valve) in its fault direction after onset. Windows are 48 steps (12 h), with onset uniform in $[10, 25]$ and the first-8-step baseline subtracted. Train, validation and test splits use alternating 7-day blocks. Training uses 4,000 windows (50/50), with half of the normal windows time-matched to positives with the fault removed. Testing uses 1,000 windows per construct and up to 300 windows per other fault file. *Clean mode* applies the fault effect only to signature channels. *Real mode* keeps every channel's real deviation.

Models. Window statistics: mean, min, max, s.d., slope and last-8 mean per channel. Logistic regression ($C = 1$, standardized), random forest (200 trees, min leaf 2), gradient boosting (scikit-learn defaults), MLP on every 4th step (hidden $128, 64$, Adam, early stopping). LSTM: 32 hidden units, BPTT, Adam (3×10^{-3}), batch 64, gradient clipping 5, up to 30 epochs, early stopping on 15% validation loss with patience 6. Channel-independent model: shared per-channel encoder

$z_c = \mathrm{relu}(x_c W_1 + b_1)$ (32 units) with a linear head $\sum_c A_c^\top z_c$. Capped CI: $\beta \sum_c \tanh(A_c^\top z_c + d_c)$, with the same encoder and a learnable β initialized at 3. All neural models are implemented in NumPy and pass finite-difference gradient checks. *Reference rule*: each signature channel minus a ridge prediction from the non-signature channels (fit on normal windows), scaled by its residual s.d., and $\max_\tau \min_{j \in S}$ of the 8-step window means after step 8. Calibrated like the models.

Remedies. Each fix adds $n_{\mathrm{train}}/2 = 2{,}000$ label-0 windows: channel near-misses (all-but-one, cycling over the left-out channel); fully staggered (half "consecutive segments", half "separate short dips"); CIS (half channel near-misses, half pair-staggered, cycling over pairs); generic negatives (the fault's effect moved onto k unrelated channels). The recipe sweep adds $c \in \{0, 3, 10, 30, 100, 300, 1000\}$ near-misses per type to an otherwise identical training set. The signal sweep multiplies D by a factor in both training and test.

Compute. All experiments run on a 2-core CPU. The synthetic suite takes about 30 minutes, one toy seed about 10 minutes, and one real-data target 10–60 minutes.

C Additional results

Table 3: Synthetic law check (S1; mean±s.d., 5 seeds). *Law*: corollary 3 evaluated at the model's own measured detection rate (no fitting).

Model	δ/σ	Detection	2-of-3	law	1-of-3	law
GD logistic	0.5	0.070±0.005	0.040±0.003	0.039	0.021±0.002	0.021
	1.0	0.274±0.012	0.122±0.007	0.120	0.041±0.003	0.040
	1.5	0.606±0.015	0.278±0.012	0.276	0.074±0.004	0.072
	2.0	0.869±0.011	0.491±0.017	0.489	0.123±0.007	0.120
	2.5	0.974±0.002	0.706±0.013	0.699	0.190±0.009	0.184
	3.0	0.998±0.000	0.863±0.006	0.867	0.273±0.010	0.272
GD 2-layer ReLU	0.5	0.026±0.004	0.019±0.002	0.019	0.014±0.001	0.014
	1.0	0.113±0.007	0.059±0.005	0.057	0.027±0.002	0.025
	1.5	0.379±0.026	0.164±0.009	0.163	0.050±0.002	0.049
	2.0	0.760±0.012	0.370±0.009	0.381	0.093±0.004	0.094
	2.5	0.941±0.004	0.582±0.022	0.605	0.141±0.009	0.151
	3.0	0.991±0.003	0.785±0.015	0.793	0.222±0.009	0.226

Table 4: Composite null (C3, $k = 3$, 4×10^5 draws). *Size*: worst false-alarm rate over look-alikes, with shifts bounded by δ or unbounded (50δ). Additive detectors are calibrated either on normal data or for the bounded null (theorem 4(c)).

	Additive, normal-calibrated			Additive, bounded-null		Intersection-union		Bound
δ/σ	size ($\leq \delta$)	size (∞)	power	size (∞)	power	size (∞)	power (pred.)	(a)
2	0.493	1.000	0.872	1.000	0.121	0.010	0.051 (0.052)	0.372
3	0.872	1.000	0.998	1.000	0.276	0.010	0.423 (0.421)	0.750
4	0.989	1.000	1.000	1.000	0.493	0.010	0.866 (0.865)	0.953
5	1.000	1.000	1.000	1.000	0.712	0.010	0.989 (0.989)	0.996

D Reproducibility

All code (data loaders, splicing, models, theory checks, plotting) is NumPy/scikit-learn based and will be released. The real-data benchmark is public (Granderson et al., 2023). Every number in section 6 is produced by the scripts named above from fixed seeds. Real-data numbers come from single runs, reported as produced.

Table 5: Separated regime (S2; mean±s.d., 3 seeds). Left block: normal-vs-target training. *Position*: $p(x^{(j)})$ of channel near-misses at the prototypes. *Bound*: $1-(R/R_{\min})/\sqrt{k}$, a valid lower bound on the position because $\kappa \le R/R_{\min}$ (lemma 10). *Flagged*: near-miss flag rate at 1% false alarms. Right block: retrained with 100 near-misses per type. *Norm ratio*: R/G after over R/G before. theorem 5(b) requires $R/R_{\min} \ge k\gamma/G$ with γ the smallest target-minus-near-miss score gap; it held in every run. Detection after retraining is 1.000 throughout.

k	linear $\frac{k-1}{k}$	position	bound	$R/R_{\min}$	flagged	norm ratio	$R/R_{\min}$	$k\gamma/G$	flagged
2	0.500	0.565±0.049	0.049	1.34	0.980	3.00±0.34	4.03	1.89	0.203
3	0.667	0.727±0.009	0.267	1.27	1.000	3.82±0.53	4.85	2.24	0.600
4	0.750	0.808±0.023	0.404	1.19	1.000	5.96±0.84	7.11	2.77	0.797
6	0.833	0.886±0.010	0.505	1.21	1.000	7.05±1.14	8.56	3.53	0.978
8	0.875	0.911±0.004	0.574	1.20	1.000	5.81±0.78	7.00	1.25	1.000

Table 6: Recipe curves: 2-of-3 look-alike flag rate vs. channel near-misses per type. Top: synthetic two-layer ReLU (S4; mean±s.d., 3 seeds). Bottom: real data, LSTM, one seed (detection ≥ 0.99 throughout).

Synthetic, per type	0	4	16	64	256
weight decay 0	0.87±0.03	0.53±0.03	0.48±0.17	0.44±0.08	0.34±0.03
weight decay 0.001	0.93±0.00	0.90±0.01	0.77±0.05	0.60±0.05	0.46±0.03

Real, per type	0	3	10	30	100	300	1000
FCU fan (clean), wd 10^{-3}	0.96	0.89	0.75	0.57	0.26	0.06	0.01
FCU fan (clean), wd 0	0.95	0.92	0.82	0.62	0.54	0.44	0.08
FCU fan (real), wd 10^{-3}	1.00	1.00	0.69	0.65	0.41	0.14	0.00
FCU fan (real), wd 0	0.99	1.00	0.78	0.74	0.55	0.32	0.12
FCU valve, wd 10^{-3}	1.00	1.00	0.89	0.72	0.51	0.11	0.09
FCU valve, wd 0	1.00	1.00	1.00	0.86	0.81	0.64	0.06
AHU coil valve, wd 10^{-3}	0.81	0.78	0.84	0.68	0.60	0.49	0.50
AHU coil valve, wd 0	0.74	0.79	0.83	0.67	0.53	0.54	0.47

Table 7: Toy replica, all models, normal-vs-target training (mean±s.d., 3 seeds). Columns *law* give corollary 3 at each model's mean detection rate with equal shares ($r = m/k$).

Model	Detection	1-of-3	law	2-of-3	law	BIG1	Fully st.	Pair-st.	UP	OFF
LogReg (stats)	0.98±0.00	0.26±0.04	0.19	0.79±0.02	0.73	1.00±0.00	0.51±0.07	0.75±0.05	0.00±0.00	0.01±0.00
Random forest (stats)	0.95±0.01	0.10±0.03	0.16	0.58±0.12	0.62	0.14±0.07	0.39±0.07	0.58±0.07	0.00±0.00	0.01±0.01
Grad. boosting (stats)	0.97±0.01	0.14±0.02	0.18	0.73±0.08	0.68	0.26±0.09	0.55±0.03	0.75±0.04	0.00±0.00	0.01±0.00
MLP	0.94±0.01	0.15±0.01	0.15	0.60±0.02	0.61	0.82±0.01	0.16±0.01	0.40±0.02	0.00±0.00	0.01±0.00
LSTM	0.94±0.02	0.13±0.00	0.15	0.55±0.03	0.60	0.65±0.08	0.14±0.02	0.27±0.02	0.00±0.00	0.00±0.00
Channel-indep.	0.97±0.01	0.20±0.03	0.18	0.73±0.02	0.68	0.99±0.01	0.19±0.02	0.44±0.07	0.00±0.00	0.00±0.00
Capped CI	0.90±0.01	0.07±0.01	0.13	0.42±0.05	0.54	0.09±0.01	0.09±0.01	0.23±0.05	0.00±0.00	0.01±0.00
Reference rule	0.89±0.03	0.05±0.01	0.13	0.21±0.04	0.52	0.05±0.02	0.09±0.03	0.33±0.06	0.00±0.00	0.02±0.00

Table 8: Toy replica: 2-of-3 / pair-staggered flag rates per model and remedy (mean over 3 seeds).

Model	normal only	+ fully stag.	+ generic (OE)	+ channel NM	+ CIS
LogReg (stats)	0.79 / 0.75	0.62 / 0.13	0.79 / 0.79	0.61 / 0.39	0.59 / 0.09
Random forest (stats)	0.58 / 0.58	0.41 / 0.15	0.67 / 0.64	0.16 / 0.38	0.19 / 0.12
Grad. boosting (stats)	0.73 / 0.75	0.46 / 0.13	0.77 / 0.76	0.18 / 0.53	0.21 / 0.10
MLP	0.60 / 0.40	0.52 / 0.29	0.56 / 0.36	0.26 / 0.29	0.34 / 0.10
LSTM	0.55 / 0.27	0.40 / 0.12	0.56 / 0.26	0.15 / 0.08	0.23 / 0.04
Channel-indep.	0.73 / 0.44	0.65 / 0.11	0.73 / 0.44	0.33 / 0.23	0.39 / 0.05
Capped CI	0.42 / 0.23	0.46 / 0.07	0.47 / 0.23	0.19 / 0.20	0.27 / 0.05

Table 9: Real data (LBNL), normal-vs-target training, all models, one seed. Clean mode. *Pairwise law*: 2-of-3 rate predicted from the detection rate and the single-sensor rates (each sensor's share r_j read from its single-sensor flag rate), a test of additivity. FCU values saturate near 1; the AHU coil is informative.

Target	Model	Det.	1-of-3	2-of-3	pairwise law	BIG1	Fully st.	Pair-st.	UP	OFF
FCU fan	LogReg (stats)	1.00	0.81	1.00	1.00	0.98	0.95	0.97	0.00	0.22
	Random forest	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.33
	Grad. boosting	1.00	0.65	1.00	1.00	0.65	0.94	0.94	0.13	0.07
	MLP	0.99	0.69	0.95	1.00	0.96	0.76	0.87	0.00	0.17
	LSTM	1.00	0.79	0.97	1.00	0.98	0.85	0.92	0.48	0.61
	Channel-indep.	1.00	0.88	0.99	1.00	1.00	0.95	0.95	0.00	0.00
	Capped CI	1.00	0.96	1.00	1.00	1.00	0.99	0.98	0.41	0.14
	Reference rule	1.00	0.08	0.35	0.25	0.06	0.33	0.82	0.01	0.19
AHU coil	LogReg (stats)	1.00	0.76	1.00	1.00	1.00	0.97	1.00	0.00	0.01
	Random forest	1.00	0.66	0.98	1.00	0.81	0.86	0.89	0.05	0.01
	Grad. boosting	1.00	0.45	0.81	0.98	0.52	0.57	0.54	0.03	0.00
	MLP	1.00	0.39	0.79	0.81	0.63	0.40	0.71	0.00	0.01
	LSTM	1.00	0.38	0.78	0.63	0.58	0.30	0.73	0.00	0.01
	Channel-indep.	1.00	0.59	0.93	0.96	0.76	0.81	0.95	0.00	0.00
	Capped CI	1.00	0.47	0.96	0.98	0.41	0.31	0.50	0.01	0.06
	Reference rule	0.97	0.37	0.74	0.78	0.38	0.74	0.90	0.01	0.01

Table 10: Real never-seen faults (LSTM vs. reference rule), flag rate by the number of signature sensors the fault moves in the fault direction; one seed.

Target, detector	0 of 3	1 of 3	2 of 3	3 of 3
FCU fan (clean), LSTM	0.22	0.63	0.79	0.93
FCU fan (clean), reference rule	0.06	0.13	0.36	0.44
FCU fan (clean), LSTM + channel near-misses	0.15	0.11	0.22	0.80
FCU valve, LSTM	0.18	0.59	0.65	1.00
FCU valve, reference rule	0.05	0.06	0.23	0.19
AHU coil valve, LSTM	0.00	0.04	0.01	–